Fateme Mazdarani

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Research Interests

My research focuses on reliable, robust, and interpretable AI, particularly large language models and mathematical reasoning. I study knowledge access and verification across representations, retrieval-supported reasoning, human-centered AI, recommender systems, and scalable machine learning.

Education

Clemson University, Ph.D. in Computer Science Jan 2025 – Present

- Advisor: Dr. Carlos Toxtli, Human-AI Empowerment Lab
- Research: robust and trustworthy LLM evaluation, mathematical reasoning, and human-centered AI
- GPA: 4.0/4.0

University of Tehran, B.Sc. in Computer Engineering Aug 2018 – Jun 2023

- GPA: 3.5/4.0
- Bachelor's thesis: A Multi-Platform Voice Recognition System Activator in Persian Language

Publications and Manuscripts

Published

- F. Mazdarani, A. C. Hernandez, and C. Toxtli. **Worker-Centered AI: Transparent Explanations for Trustworthy Task Recommendation in Crowd Work.** *2026 IEEE Conference on Artificial Intelligence (CAI)*, pp. 295–302, 2026. DOI: 10.1109/CAI68641.2026.11536153.

Accepted for Publication

- F. Mazdarani and C. Toxtli. **Beyond the Answer Key: Robustness Evaluation of Large Language Models for Step-Level Mathematical Verification.** *2026 IEEE International Conference on Machine Learning and Applications (ICMLA).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **Randomized SVD Approximations for Spectral Co-Clustering of Word-Document Matrices.** *2026 IEEE International Conference on Machine Learning and Applications (ICMLA).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **TREAT: Evaluating Access to Formal Knowledge across Equivalent Mathematical Representations.** *2026 International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **The Shape of Mathematical Creativity: Measuring Mathematical Exploration in Formal Proof Generation.** *MathAI Workshop at NeurIPS 2026.* Accepted for publication.
- F. Mazdarani and C. Toxtli. **From Recognition to Reconstruction: Towards Robust Procedural Mathematical Reasoning.** *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026); Workshop on Mathematical Natural Language Processing.* Accepted for publication.

Under Review

- F. Mazdarani and C. Toxtli. **The Models Behind the Claim: Empirical Standards for LLM Evaluation.** *International Conference on Learning Representations (ICLR).* Under review.

Honors and Awards

- Dr. Robert M. Geist III Endowed Fellowship in Computing, Clemson University, 2025–2026
- Ranked 79th in the National University Entrance Exam in Iran among more than 300,000 participants, 2018
- Accepted in the First Round of the Iranian National Mathematics Olympiad, 2015–2016

Research Experience

Graduate Research Assistant Fall 2026 – Present

Human-AI Empowerment Lab, Clemson University

Research in the lab since August 2025.

- **LLM reasoning and evaluation:** develop benchmarks for theorem recognition across equivalent representations, step-level verification, and retrieval-supported procedural reasoning.
- **Autonomous underwater vehicles (AUVs):** develop and evaluate vision-language-action navigation in simulation, integrating object detection and safety checks; compare grid-based and learned controllers for search coverage, target acquisition, travel efficiency, and inference latency.
- **Worker-centered AI:** study how explanations and navigation cues affect fairness, trust, empowerment, usability, and task selection in crowd work.
- **Scalable machine learning:** investigate randomized SVD for spectral co-clustering of word-document matrices and ADMM-based sparse spectral clustering.

Teaching Experience

Graduate Teaching Assistant, Clemson University 2025 – 2026

- Served as teaching assistant for Design and Analysis of Algorithms and Human-Computer Interaction / Human-Centered Computing courses.
- Supported grading, rubric-based feedback, project checkpoints, student questions, and course assessment.

Teaching Assistant, Engineering Probability and Statistics, University of Tehran Fall 2021

Teaching Assistant, Discrete Mathematics, University of Tehran Fall 2019 – Spring 2021

Academic Service

Social Chair Summer 2026 – Present

School of Computing Graduate Student Association, Clemson University

Delegate Fall 2026 – Present

Clemson Graduate Student Government (GSG)

Peer Reviewer

Reviewed 10+ papers at various conferences.

Poster Judge August 2026

Creative Inquiry undergraduate research program, Clemson University

Skills

- **Programming Languages:** Python, MATLAB, Java, C/C++, R, SQL, Assembly (Intel x86, MIPS)
- **Machine Learning and Data Science:** PyTorch, TensorFlow, Keras, NumPy, pandas, scikit-learn, Matplotlib, statistical analysis, experiment design
- **LLM and AI Evaluation:** prompt engineering, LLM-as-a-judge evaluation, process-level evaluation, benchmark design, Hugging Face Transformers, OpenAI API, Gemini, Claude
- **Web, Scripting, and Tools:** HTML, CSS, JavaScript, React, LaTeX, Bash, Git, Docker, Maven
- **Hardware and Simulation Tools:** Arduino, ModelSim, Intel Quartus Prime, LTspice, Proteus circuit simulator

Selected Technical Projects

- **Quantum Machine Learning for Cybersecurity** (2025). Evaluated hybrid quantum deep learning and variational quantum classifiers for botnet DGA detection.
- **Sparse Spectral Clustering** (2025). Studied ADMM-based optimization methods for sparse spectral clustering.
- **Persian-Language Voice Activator** (2022–2023). Built an RNN-based voice activator with Keras and TensorFlow; bachelor's thesis, University of Tehran.
- **Movie Information Management System** (2022). Developed a full-stack application using Java, React, REST APIs, SQL, Maven, unit testing, and Tomcat.
- **CNN Acceleration by Removing Zero-Valued Neurons** (2022). Authored a course research paper on CNN acceleration through zero-valued neuron removal.
- **Smart Irrigation System** (2022). Simulated sensors and motors with Arduino IDE and Proteus for a real-time embedded system.
- **Android Surface Scanner** (2022). Developed a surface-scanning application using Java and Android Studio.
- **SIMD Instructions for IP Acceleration** (2021). Implemented acceleration techniques using C++ and Intel SIMD instructions.